Part 4

ANNOUNCEMENT

- **Deliverable 3**: April 19th, at 11:59 pm

ANNOUNCEMENT

- **Presentations – Dates**

April 22nd: Meow Meow, Hoomans, The Trinity

April 27th: Failed Experiments, Aster, Magscienticks, Trial & Error

April 29th: The Quarks, Among Us, Group #1

THE PLAN

- Recap
- Causal Graphs
    1. The Elements of Causal Graphs
    2. Representing Different Varieties of Causation
- Exercises

Indeterministic Causation

- In module 1 & 2, the causal relationships we studied were "deterministic," that is, the effect is always completely determined by the values assigned to the causes.
- ***In an "indeterministic" causal relationship, the effect is not always determined by the causal assignments.***
- Almost every causal relationship we study in the social, behavioral, and medical sciences at least appears to be indeterministic.
  - Having wealthy parents is a cause of wealth, but it does not guarantee it. Eating well and exercising regularly cause good health, but they don't guarantee it.

Indeterministic Causation

(1) Indeterministic Response Structure

- **Deterministic Response Structure**: In a *deterministic response structure*, every causal assignment completely determines the value of the effect.
- **Indeterministic Response Structure**: In an *indeterministic response structure*, there is at least one causal assignment that does not determine the value of the effect.
  - Example: Cell Phones

Indeterministic Causation

- There are *three core ideas* in this module No 3 – ***'Indeterministic Causation'***.

1. The first is that causal relations are **sometimes indeterministic.**
2. The second is that sometimes they are indeterministic in appearance only. In most cases we don't explicitly mention all of the causes that act on an effect of interest, but only some of them. If we did list all of the causes of a particular effect, then the relationship between the causes and the effect might well be deterministic. ***When we leave some out, however, the relationship will appear to be indeterministic***.
3. The third core idea is this: even though a particular causal assignment might not always determine whether a given effect will or will not occur, ***it can determine the chances, or the probability that the effect will occur***.

Indeterministic Causation

- <u>Pseudo-indeterminism vs. True Indeterminism</u>

  - **Pseudo-indeterministic Response Structure** - A response structure for an effect E is pseudo-indeterministic when i) it is an indeterministic response structure for E, and ii) there are other causes of E that, when added, make the response structure deterministic.

  - **Truly Indeterministic Response Structure** - A response structure for an effect E is truly-indeterministic when i) it is an indeterministic response structure for E, and ii) there is no set of causes we can add that will make it a deterministic response structure.

Indeterministic Causation

- <u>(2) Sources of Indeterminism</u>

- **Response Structure Uniformity** - a population has response structure uniformity for a given effect if every individual in the population has the same response structure for that effect.

- **Indeterministic Causation in a Population** – if there are two individuals in a population who exhibit different effects even though they have the same causal assignment, then there is *apparent* **indeterministic causation in the population**. Causation in a population can appear indeterministic for two reasons:

  1. Some individual in the population has an indeterministic response structure.
  2. Every individual in the population has a deterministic response structure, but the population is not response structure uniform.

Indeterministic Causation

- <u>(5) Definition of Indeterministic Causation</u>

  ▪ What is it for the effect to change, however? Does it mean that the effect must take on a different value in the two causal assignments? Or is it enough that the chances for the effect change?

- Clearly, in the case of indeterministic causation, only the chances need to change. So let's modify the definition of cause ever so slightly to accommodate indeterminism:

  ▪ **Direct (Indeterministic) Cause In a system of variables S**, X is a direct cause of Y relative to S if and only if there is at least one test pair of causal assignments for X in the response structure Y across which the probability distribution over Y differs.

- *The "probability distribution" means, roughly, the probability given to each of the possible values Y can take on. If the probability given to any of Y's values changes, then the probability distribution over Y has changed.*

Last time!

Indeterministic Causation

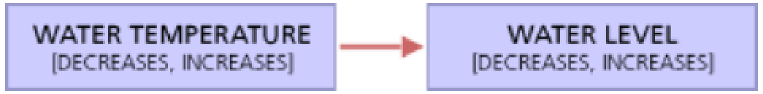
- <u>(6) Direct vs. Indirect Causation</u>
  - If X is a direct cause of Y relative to some causal system S, then we can hold the other variables in S constant and produce a change in Y as a result of changing X.
  - If X is only an indirect cause of Y, however, then all of X's influence on Y must go through some other variable in S, say V. X must first produce a change in V, which must in turn produce a change Y.
  - In the response structure of Y, however, any test pair of X will involve two assignments in which V does not change, and so X will not appear as a direct cause of Y. That does *not* mean that X is not a cause of Y, it just means that it is not a *direct* cause of Y.

Reading Exercise – No 16 (Module 4 | Causal Graphs)

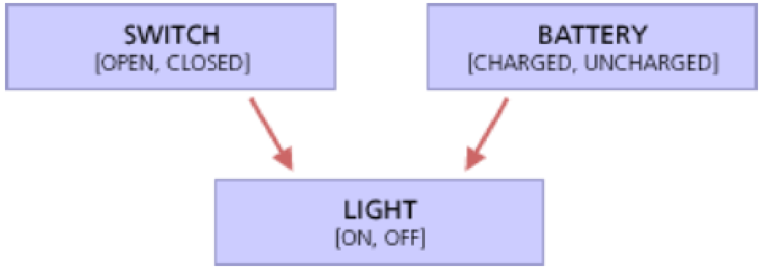
In previous modules, we built up an account of what it is for one variable X to be a direct cause of another Y.

Note - The set of values for a variable must be both exclusive and exhaustive.

Module 4 – discusses a compact, pictorial way to represent the causal relations, direct and indirect, among a system of variables.



The causal relations between the state of a light bulb, a light switch and a battery can be represented by a causal graph involving three variables:

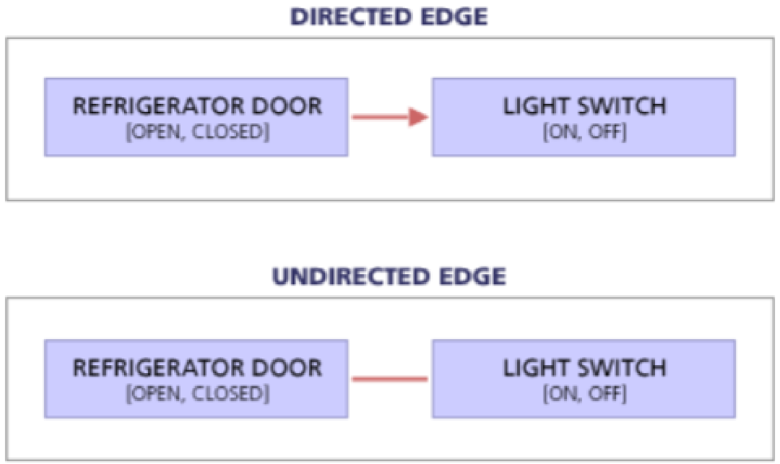


and the relations between the refrigerator door, the light switch, and the refrigerator light by another:

CAUSAL GRAPHS

- <u>1. The Elements of Causal Graphs</u>
  - Causal Graphs represent the causal relations in a causal system. Specifically, causal graphs involve:
    a) A set of variables, and
    b) A set of directed edges that connect the variables.

A direct edge in a causal graph is an arrow, where the *head* of the arrow points to the effect variable and the *tail* comes from the causa variable.



We include a *directed edge* from a variable X to a variable **Y** in the causal graph that represents a set of variables S *if and only* if **X** is a *direct cause* of relative to S.

CAUSAL GRAPHS

- 1. The Elements of Causal Graphs
  - Causal Graphs represent the causal relations in a causal syste graphs involve:
    a) A set of variables, and
    b) A set of directed edges that connect the variables.

The edges are "directed" because causation is asymmetric and has a direction.

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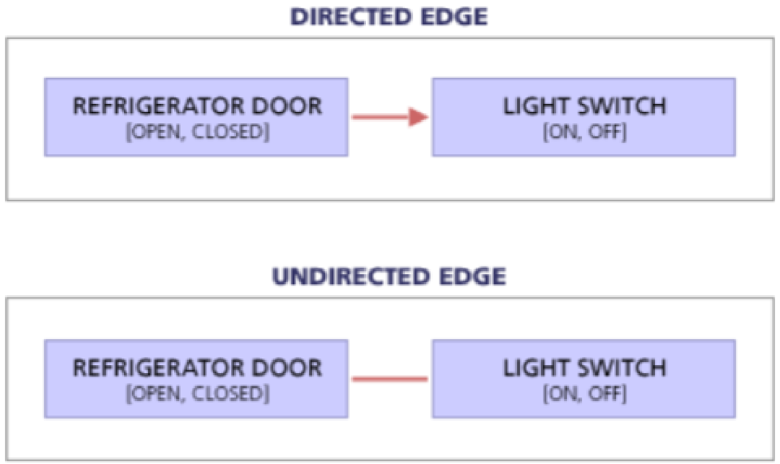

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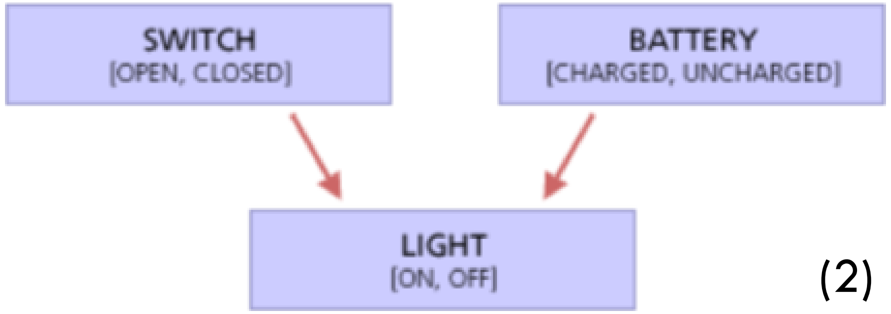
CAUSAL GRAPHS

Example: Switch and Lights



Note:

(1) There is *no* arrow from the switch to the battery, nor from the battery to the switch, even though they are physically connected by wire on the circuit.

- **WHY?**
- Because the state of the battery has no causal influence on the state of the switch, nor does the state of the switch have any direct influence on the state of the battery. Intervening to change the state of the battery will not affect the state of the switch, even though the two are physically connected.

(2) There *is* an arrow from the switch to the light bulb, even though, when the battery is uncharged, changing the switch from open to closed (or from closed to open) will not change the state of the light bulb (it will stay off).

(3) There is an arrow from the switch to the light bulb because there is *some* state of the battery, namely when it is charged, for which changing the causal assignment of the switch does change the state of the light.

CAUSAL GRAPHS

- 2. Representing Different Varieties of Causation
- *Recall:*
  - A variable C is a common cause of two or more other variables X and Y when C is a cause (direct or indirect) of both X and Y.

    - Direct Cause: If, in a system of variables S there are any test pair of causal assignments for X in which there is a difference in the effect Y, then X is a *direct cause* of Y relative to S.

    - Indirect Cause: If X is only an indirect cause of Y, however, then all of X's influence on Y must go through some other variable in S, say V. X must first produce a change in V, which must in turn produce a change in Y.

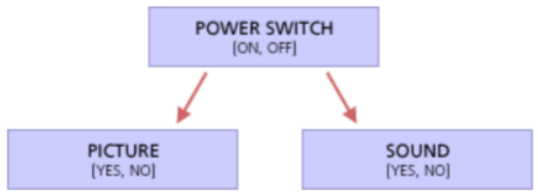
CAUSAL GRAPHS

- <u>2. Representing Different Varieties of Causation</u>
- *Recall:*
  - A variable C is a common cause of two or more other variables X and Y when C is a cause (direct or indirect) of both X and Y.

Variables For A Tv System

Variable	Value
SOUND	[Yes, No]
POWER SWITCH	[On, Off]
PICTURE	[Yes, No]

The causal graph for these variables is:



Here POWER SWITCH is a common cause of both PICTURE and SOUND because changing the state of POWER SWITCH changes both the value of PICTURE and the value of SOUND.

CAUSAL GRAPHS

- <u>2. Representing Different Varieties of Causation</u>
  - If one variable only influences another through some intermediate variable, then there is *no* arrow between the first feature and the third feature in the chain.

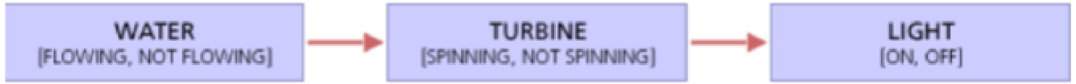
For example, in the scenario involving a light be powered by hydro-electricity, the amount of water coming from the damn through the spout influences the speed of the turbine which influences the speed of the turbine which influences whether electricity is generated to power the light bulb:

The water is still a cause of the light bulb, but only an indirect one. If the variables and their values are:

Variables For Turbine System

Variable	Value
WATER	[Flowing, Not flowing]
TURBINE	[Spinning, Not spinning]
LIGHT	[On, Off]

then the causal graph is as follows:



CAUSAL GRAPHS

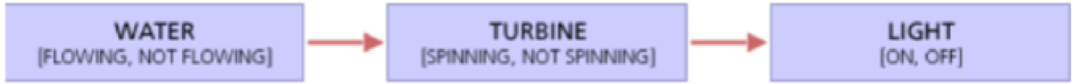
- <u>2. Representing Different Varieties of Causation</u>
  - If one variable only influences another through some intermediate variable, then there is *no* arrow between the first feature and the third feature in the chain.

For example, in the scenario involving a light be powered by hydro-electricity, the amount of water coming from the damn through the spout influences the speed of the turbine which influences the speed of the turbine which influences whether electricity is generated to power the light bulb:

The water is still a cause of the light bu ... alues are:

Note: If there is a chain of arrows from A to C, i.e., A -> B -> C, etc., then we say there is a ***causal chain*** from A to C.

then the causal graph is as follows:



CAUSAL GRAPHS

- 2. Representing Different Varieties of Causation
  - Causation among variables is asymmetric, but it isn't antisymmetric.
    - A relationship is antisymmetric if the fact that it holds one way precludes it holding the other. For example, the relationship "is a parent of" is antisymmetric. If person X is a parent of Y, then Y cannot be a parent of X.

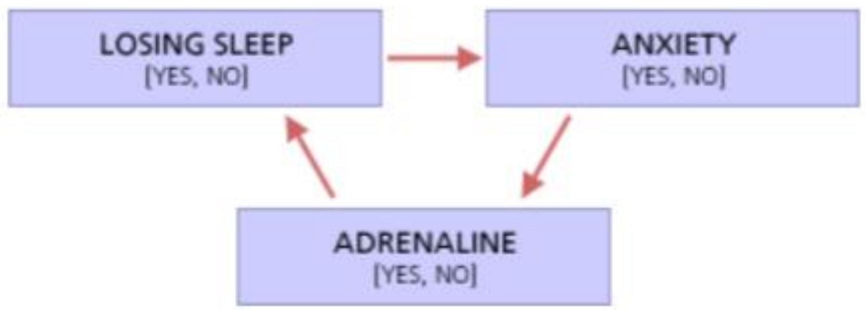
CAUSAL GRAPHS

- <u>2. Representing Different Varieties of Causation</u>
  - It is possible for one variable X to be a cause of Y and also for Y to be a cause of X.

In each of these cases, we say that there is a ***direct cycle*** in the causal graph.

CAUSAL GRAPHS

- <u>2. Representing Different Varieties of Causation</u>
  - Cycles of Causality need not be direct.



If there is a chain of edges leading from any variable back to itself, then we say the graph has a ***cycle.***

If the graph has a cycle, we say it is a cyclic graph.

Reading Exercise – No 16 (Module 4 | Causal Graphs)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

A - Reflect and discuss – Causal Graphs – Variables 1

Question No 01 – Causal graphs represent: (choose one)

A. Causal relationships between particular events.

B. Causal relationships between test pairs

C. Causal relationships between variables.

D. None of the above.

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A - Reflect and discuss – Causal Graphs – Variables 1

Question No 02 – Which of the following does a causal graph contain? (choose one)

A. Causal assignments and test pairs.

B. The probability distribution over the variable designated as the effect.

C. Variables and directed edges.

D. A response structure.

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A - Reflect and discuss – Causal Graphs – Variables 1

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Reading Exercise – No 16 (Module 4 | Causal Graphs)

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B - Reflect and discuss – Causal Graphs – Directed Edges

Question No 03 – A directed edge from X to Y in a causal graph means that: (choose one)

A. X promotes the occurrence of Y, relative to the other variables in the system.
B. Y always occurs when X does, relative to the other variables in the system.
C. X is the only cause of Y, relative to the other variables in the system.
D. X is a direct cause of Y, relative to the other variables in the system.

Reading Exercise – No 16 (Module 4 | Causal Graphs)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

B - Reflect and discuss – Causal Graphs – Directed Edges

Question No 03 – A directed edge from X to Y in a causal graph means that: (choose one)

A. X promotes the occurrence of Y, relative to the other variables in the system.

B. Y always occurs when X does, relative to the other variables in the system.

C. X is the only cause of Y, relative to the other variables in the system.

D. X is a direct cause of Y, relative to the other variables in the system.

Answer – D

Directed edges can also represent preventative causes. For example, the variables might be Regular Exercise and Heart Disease.

Reading Exercise – No 16 (Module 4 | Causal Graphs)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

B - Reflect and discuss – Causal Graphs – Directed Edges

Question No 04 – If there is a directed edge from X to Y in a causal graph, then this means that:

A. Whenever X occurs, Y must occur also.

B. There is at least one test pair of causal assignments for X in which there is a difference in Y.

C. Y cannot occur except when X is present.

D. Y is a direct cause of X.

Reading Exercise – No 16 (Module 4 | Causal Graphs)

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B - Reflect and discuss – Causal Graphs – Directed Edges

Question No 04 – If there is a directed edge from X to Y in a causal graph, then this means that:

A. Whenever X occurs, Y must occur also.

B. There is at least one test pair of causal assignments for X in which there is a difference in Y.

C. Y cannot occur except when X is present.

D. Y is a direct cause of X.

Answer – B

Please note that X might be an indeterministic cause of Y.

Reading Exercise – No 16 (Module 4 | Causal Graphs)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

C - Reflect and discuss – Deterministic Causation: The Malaria Example

Question No 05 – Referring to the response structure in the table titled: Response Structure For The Malaria Case, which of the following are test pairs of causal assignments for Inoculated? (Note: Not all test pairs are on this list.) (choose all that apply)

A. Assignments 1 and 3.

B. Assignments 2 and 6.

C. Assignments 4 and 7.

D. Assignments 9 and 13.

Response Structure For The Malaria Case

Assignment	Variable 1: BITTEN	Variable 2: INOCULATED	Variable 3: HAS GENE	Variable 4: DRINKER	Effect: MALARIA
1	True	True	True	True	False
2	True	True	True	False	False
3	True	True	False	True	False
4	True	True	False	False	False
5	True	False	True	True	False
6	True	False	True	False	False
7	True	False	False	True	True
8	True	False	False	False	True
9	False	True	True	True	False
10	False	True	True	False	False
11	False	True	False	True	False
12	False	True	False	False	False
13	False	False	True	True	False
14	False	False	True	False	False
15	False	False	False	True	False
16	False	False	False	False	False

Reading Exercise – No 16 (Module 4 | Causal Graphs)

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C - Reflect and discuss – Deterministic Causation: The Malaria Example

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Response Structure For The Malaria Case

Assignment	Variable 1: BITTEN	Variable 2: INOCULATED	Variable 3: HAS GENE	Variable 4: DRINKER	Effect: MALARIA
1	True	True	True	True	False
2	True	True	True	False	False
3	True	True	False	True	False
4	True	True	False	False	False
5	True	False	True	True	False
6	True	False	True	False	False
7	True	False	False	True	True
8	True	False	False	False	True
9	False	True	True	True	False
10	False	True	True	False	False
11	False	True	False	True	False
12	False	True	False	False	False
13	False	False	True	True	False
14	False	False	True	False	False
15	False	False	False	True	False
16	False	False	False	False	False

Answer – B and D

Reading Exercise – No 16 (Module 4 | Causal Graphs)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

C - Reflect and discuss – Deterministic Causation: The Malaria Example

Question No 06 – If we built a causal graph of the system that includes: {Bitten, Inoculated, Gene, Drinker, Malaria}, based on the response structure in the table titled: Response Structure For The Malaria Case, **should there be a directed edge from 'Inoculated' to 'Malaria'? (choose one)**

A. Yes

B. No

Response Structure For The Malaria Case

Assignment	Variable 1: BITTEN	Variable 2: INOCULATED	Variable 3: HAS GENE	Variable 4: DRINKER	Effect: MALARIA
1	True	True	True	True	False
2	True	True	True	False	False
3	True	True	False	True	False
4	True	True	False	False	False
5	True	False	True	True	False
6	True	False	True	False	False
7	True	False	False	True	True
8	True	False	False	False	True
9	False	True	True	True	False
10	False	True	True	False	False
11	False	True	False	True	False
12	False	True	False	False	False
13	False	False	True	True	False
14	False	False	True	False	False
15	False	False	False	True	False
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Reading Exercise – No 16 (Module 4 | Causal Graphs)

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C - Reflect and discuss – Deterministic Causation: The Malaria Example

Question No 06 – If we built a causal graph of the system that includes: {Bitten, Inoculated, Gene, Drinker, Malaria}, based on the response structure in the table titled: Response Structure For The Malaria Case, **should there be a directed edge from 'Inoculated' to 'Malaria'? (choose one)**

A. Yes

B. No

Response Structure For The Malaria Case

Assignment	Variable 1: BITTEN	Variable 2: INOCULATED	Variable 3: HAS GENE	Variable 4: DRINKER	Effect: MALARIA
1	True	True	True	True	False
2	True	True	True	False	False
3	True	True	False	True	False
4	True	True	False	False	False
5	True	False	True	True	False
6	True	False	True	False	False
7	True	False	False	True	True
8	True	False	False	False	True
9	False	True	True	True	False
10	False	True	True	False	False
11	False	True	False	True	False
12	False	True	False	False	False
13	False	False	True	True	False
14	False	False	True	False	False
15	False	False	False	True	False
16	False	False	False	False	False

Answer - A.

Try looking at the test pair 3 and 7, or the test pair 4 and 8. In each of these pairs, Inoculated is the only variable that changes, and Malaria also changes values. Therefore, Inoculated is a direct cause of Malaria and the causal graph representing this system should have an edge from Inoculated to Malaria.

Reading Exercise – No 16 (Module 4 | Causal Graphs)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

C - Reflect and discuss – Deterministic Causation: The Malaria Example

Question No 07 – Which of the following pairs of causal assignments should we compare to determine whether Drinker is a direct cause of Malaria? (choose all that apply)

A. Assignments 1 and 2.

B. Assignments 15 and 16.

C. Assignments 5 and 7.

D. Assignments 11 and 12.

Response Structure For The Malaria Case

Assignment	Variable 1: BITTEN	Variable 2: INOCULATED	Variable 3: HAS GENE	Variable 4: DRINKER	Effect: MALARIA
1	True	True	True	True	False
2	True	True	True	False	False
3	True	True	False	True	False
4	True	True	False	False	False
5	True	False	True	True	False
6	True	False	True	False	False
7	True	False	False	True	True
8	True	False	False	False	True
9	False	True	True	True	False
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12	False	True	False	False	False
13	False	False	True	True	False
14	False	False	True	False	False
15	False	False	False	True	False
16	False	False	False	False	False

Reading Exercise – No 16 (Module 4 | Causal Graphs)

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C - Reflect and discuss – Deterministic Causation: The Malaria Example

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C. Assignments 5 and 7.

D. Assignments 11 and 12.

Response Structure For The Malaria Case

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3	True	True	False	True	False
4	True	True	False	False	False
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6	True	False	True	False	False
7	True	False	False	True	True
8	True	False	False	False	True
9	False	True	True	True	False
10	False	True	True	False	False
11	False	True	False	True	False
12	False	True	False	False	False
13	False	False	True	True	False
14	False	False	True	False	False
15	False	False	False	True	False
16	False	False	False	False	False

Answer – A, B and D.

Note - Remember that we're trying to determine whether Drinker is a direct cause of Malaria.

Reading Exercise – No 16 (Module 4 | Causal Graphs)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

C - Reflect and discuss – Deterministic Causation: The Malaria Example

Question No 08 – If we built a causal graph of the system that includes: {Bitten, Inoculated, Gene, Drinker, Malaria}, **should there be a directed edge from 'Drinker' to 'Malaria'? (choose one)**

A. Yes

B. No

Response Structure For The Malaria Case

Assignment	Variable 1: BITTEN	Variable 2: INOCULATED	Variable 3: HAS GENE	Variable 4: DRINKER	Effect: MALARIA
1	True	True	True	True	False
2	True	True	True	False	False
3	True	True	False	True	False
4	True	True	False	False	False
5	True	False	True	True	False
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10	False	True	True	False	False
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Reading Exercise – No 16 (Module 4 | Causal Graphs)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

C - Reflect and discuss – Deterministic Causation: The Malaria Example

Question No 08 – If we built a causal graph of the system that includes: {Bitten, Inoculated, Gene, Drinker, Malaria}, **should there be a directed edge from 'Drinker' to 'Malaria'? (choose one)**

A. Yes

B. No

Response Structure For The Malaria Case

Assignment	Variable 1: BITTEN	Variable 2: INOCULATED	Variable 3: HAS GENE	Variable 4: DRINKER	Effect: MALARIA
1	True	True	True	True	False
2	True	True	True	False	False
3	True	True	False	True	False
4	True	True	False	False	False
5	True	False	True	True	False
6	True	False	True	False	False
7	True	False	False	True	True
8	True	False	False	False	True
9	False	True	True	True	False
10	False	True	True	False	False
11	False	True	False	True	False
12	False	True	False	False	False
13	False	False	True	True	False
14	False	False	True	False	False
15	False	False	False	True	False
16	False	False	False	False	False

Answer - B.

Remember that we only include an edge from Drinker to Malaria if there is at least one test pair of causal assignments for Drinker across which Malaria varies. Are there any such pairs of causal assignments?

Reading Exercise – No 16 (Module 4 | Causal Graphs)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

D - Reflect and discuss – Indeterministic Causation: The Cell Phone Example 1

Question No 09 – Referring to the response structure for CONNECTED shown in the table titled: Deterministic Response Structure For Connected, **which of the following are test pairs for CALL PLACED? (choose all that apply)**

A. 1 and 2.

B. 1 and 3.

C. 1 and 4.

D. 2 and 3.

E. 2 and 4.

F. 3 and 4.

Deterministic Response Structure For Connected

Assignment	CALL PLACED	IN RANGE OF TOWER	CONNECTED
1	Send	Yes	Yes
2	Send	No	No
3	End	Yes	No
4	End	No	No

Reading Exercise – No 16 (Module 4 | Causal Graphs)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

D - Reflect and discuss – Indeterministic Causation: The Cell Phone Example 1

Question No 09 – Referring to the response structure for CONNECTED shown in the table titled: Deterministic Response Structure For Connected, **which of the following are test pairs for CALL PLACED? (choose all that apply)**

A. 1 and 2.
B. 1 and 3.
C. 1 and 4.
D. 2 and 3.
E. 2 and 4.
F. 3 and 4.

Deterministic Response Structure For Connected

Assignment	CALL PLACED	IN RANGE OF TOWER	CONNECTED
1	Send	Yes	Yes
2	Send	No	No
3	End	Yes	No
4	End	No	No

Answer – B and E

Reading Exercise – No 16 (Module 4 | Causal Graphs)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

D - Reflect and discuss – Indeterministic Causation: The Cell Phone Example 1

Question No 10 – In which of the following test pairs for CALL PLACED does CALL CONNECTED vary? (choose one)

A. 1 and 3

B. 2 and 4

C. Neither.

Deterministic Response Structure For Connected

Assignment	CALL PLACED	IN RANGE OF TOWER	CONNECTED
1	Send	Yes	Yes
2	Send	No	No
3	End	Yes	No
4	End	No	No

Reading Exercise – No 16 (Module 4 | Causal Graphs)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

D - Reflect and discuss – Indeterministic Causation: The Cell Phone Example 1

Question No 10 – In which of the following test pairs for CALL PLACED does CALL CONNECTED vary? (choose one)

A. **1 and 3**

B. 2 and 4

C. Neither.

Deterministic Response Structure For Connected

Assignment	CALL PLACED	IN RANGE OF TOWER	CONNECTED
1	Send	Yes	Yes
2	Send	No	No
3	End	Yes	No
4	End	No	No

Reading Exercise – No 16 (Module 4 | Causal Graphs)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

E - Reflect and discuss – Indeterministic Causation: The Cell Phone Example 2

Question No 11 – If we consider the causal system: {Call Placed, Battery Charged, In Tower Range, Connected}, with the response structure given, Call Placed is a direct cause of Call Connected. **If we consider the smaller system including only: {Call Placed, Battery Charged, Connected}, the response structure for Connected would be: (choose all that apply)**

A. Deterministic.
B. Truly-Indeterministic.
C. Pseudo-Indeterministic.
D. None of the above.

Assignment	Call Placed	Battery Charged	In Range of Tower	Effect: Call Connected
1	Send	Yes	Yes	Yes
2	Send	Yes	No	No
3	Send	No	Yes	No
4	Send	No	No	No
5	End	Yes	Yes	No
6	End	Yes	No	No
7	End	No	Yes	No
8	End	No	No	No

Reading Exercise – No 16 (Module 4 | Causal Graphs)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

E - Reflect and discuss – Indeterministic Causation: The Cell Phone Example 2

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A. Deterministic.
B. Truly-Indeterministic.
C. Pseudo-Indeterministic.
D. None of the above.

Assignment	Call Placed	Battery Charged	In Range of Tower	Effect: Call Connected
1	Send	Yes	Yes	Yes
2	Send	Yes	No	No
3	Send	No	Yes	No
4	Send	No	No	No
5	End	Yes	Yes	No
6	End	Yes	No	No
7	End	No	Yes	No
8	End	No	No	No

Answer - C.

When we include the variable In Tower Range, as above, we have a deterministic response structure, thus the reduced system is pseudo-indeterministic, not deterministic or truly-indeterministic. The response structure is no longer deterministic after we remove the variable ‘In Tower Range’, because the causal assignment: ‘Call Placed = Send’, ‘Battery Charged = Yes’ does not determine if the call gets through. If In Tower Range happens to be Yes, the call gets through. But if it happens to be No, then the call doesn’t.

Reading Exercise – No 16 (Module 4 | Causal Graphs)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

E - Reflect and discuss – Indeterministic Causation: The Cell Phone Example 2

Question No 12 – Again, relative to the smaller system including only: {Call Placed, Battery Charged, Connected}, Call Placed is a direct cause of Connected if: (choose all that apply)

A. There is at least one test pair for Call Placed such that Connected changes from Yes to No, or from No to Yes.

B. There is at least one test pair for Call Placed such that the probability of Connected changes.

C. Neither of the above.

Assignment	Call Placed	Battery Charged	In Range of Tower	Effect: Call Connected
1	Send	Yes	Yes	Yes
2	Send	Yes	No	No
3	Send	No	Yes	No
4	Send	No	No	No
5	End	Yes	Yes	No
6	End	Yes	No	No
7	End	No	Yes	No
8	End	No	No	No

Reading Exercise – No 16 (Module 4 | Causal Graphs)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

E - Reflect and discuss – Indeterministic Causation: The Cell Phone Example 2

Question No 12 – Again, relative to the smaller system including only: {Call Placed, Battery Charged, Connected}, Call Placed is a direct cause of Connected if: (choose all that apply)

A. **There is at least one test pair for Call Placed such that Connected changes from Yes to No, or from No to Yes.**

B. **There is at least one test pair for Call Placed such that the probability of Connected changes.**

C. Neither of the above.

Assignment	Call Placed	Battery Charged	In Range of Tower	Effect: Call Connected
1	Send	Yes	Yes	Yes
2	Send	Yes	No	No
3	Send	No	Yes	No
4	Send	No	No	No
5	End	Yes	Yes	No
6	End	Yes	No	No
7	End	No	Yes	No
8	End	No	No	No

Answer - A and B.

If there is a test pair for Call Placed such that the probability of Connected changes, then Call Placed is an indeterministic direct cause of Connected.

Reading Exercise – No 16 (Module 4 | Causal Graphs)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

F - Reflect and discuss – Causal Chains 1

Question No 13 – What is the difference between a causal system in which there is a causal chain from x to y to z but in which x is not a direct cause of z, and another causal system in which there is a causal chain from x to y to z but in which x is also a direct cause of z?

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

F - Reflect and discuss – Causal Chains 1

Question No 13 – What is the difference between a causal system in which there is a causal chain from x to y to z but in which x is not a direct cause of z, and another causal system in which there is a causal chain from x to y to z but in which x is also a direct cause of z?

Answer - The difference cannot be seen in experiments in which only x is wiggled and z is observed. In both of these systems, wiggling x will have an effect on y and z. The difference shows up when we consider what would happen if we intervened to hold y fixed and wiggled x. In the first system, we would observe no change in z, because all of the influence of x on z goes through y. In the second system, however, we would still observe a change in z because x influences z both indirectly through y (a route we have blocked by holding y fixed), but also directly.

The causal system involving variables Education, Income, and Happiness, is an example of the second type of system. Education tends to affect Income, which tends to affect Happiness, but Education can also affect Happiness directly through the non-monetary enrichment it provides in an individual’s life, even in cases where Education does not have any effect on Income.

Reading Exercise – No 16 (Module 4 | Causal Graphs)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

G - Reflect and discuss – Causal Chains 2

Question No 14 – Consider the causal system: {Exposure (to other kids with Chicken Pox), Infected (with Chicken Pox Virus), Have Chicken Pox Rash}. **Using your common sense, is Exposure a direct cause of Have Chicken Pox Rash relative to this system? (choose one)**

A. Yes

B. No

C. Not enough information to tell

Reading Exercise – No 16 (Module 4 | Causal Graphs)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

G - Reflect and discuss – Causal Chains 2

Question No 14 – Consider the causal system: {Exposure (to other kids with Chicken Pox), Infected (with Chicken Pox Virus), Have Chicken Pox Rash}. **Using your common sense, is Exposure a direct cause of Have Chicken Pox Rash relative to this system? (choose one)**

A. Yes

B. No

C. Not enough information to tell

Reading Exercise – No 16 (Module 4 | Causal Graphs)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

G - Reflect and discuss – Causal Chains 2

Question No 15 – Consider the smaller system: {Exposure (to other kids with Chicken Pox), Have Chicken Pox Rash}. **Using your common sense, is Exposure a direct cause of Have Chicken Pox Rash relative to this system? (choose one)**

A. Yes

B. No

Reading Exercise – No 16 (Module 4 | Causal Graphs)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

G - Reflect and discuss – Causal Chains 2

Question No 15 – Consider the smaller system: {Exposure (to other kids with Chicken Pox), Have Chicken Pox Rash}. **Using your common sense, is Exposure a direct cause of Have Chicken Pox Rash relative to this system? (choose one)**

A. **Yes**

B. No

Answer – A

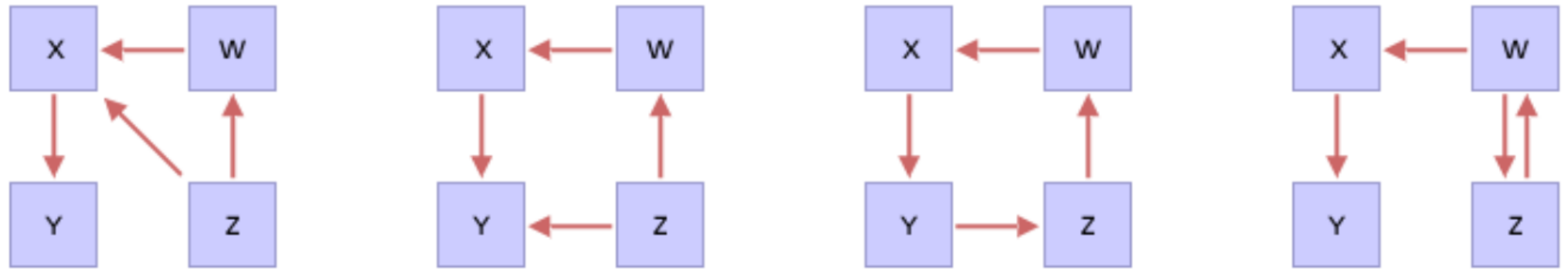
If we intervened to change a child's exposure, then we would change the probability of their Having ChickenvPox Rash. Thus, in this small system, Exposure is a direct cause.

Reading Exercise – No 16 (Module 4 | Causal Graphs)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

H - Reflect and discuss – Cyclic Causal Graphs

Question No 16 – Which of the following causal graphs contains direct cycles? (choose all that apply)



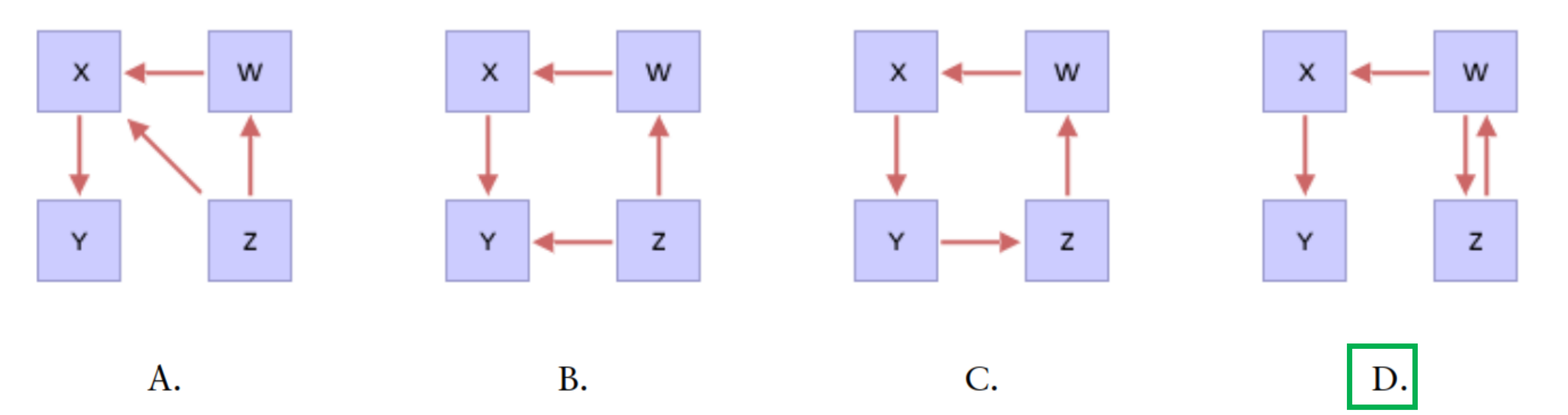
A. B. C. D.

Reading Exercise – No 16 (Module 4 | Causal Graphs)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

H - Reflect and discuss – Cyclic Causal Graphs

Question No 16 – Which of the following causal graphs contains direct cycles? (choose all that apply)



Answer - D.

Graph A does not have a direct cycle; there are no two variables that are each causes of the other.
Graph B does not have a direct cycle; there are no two variables that are each causes of the other.
Graph C has a cycle, but it isn't a direct cycle.

Note - a direct cycle is just when we have two variables, X and Y, and X is a direct cause of Y, and Y is also a direct cause of X.

Reading Exercise – No 16 (Module 4 | Causal Graphs)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

I - Reflect and discuss – Case Study 1

Question No 17 – Consider the following list:

1. Adult [yes, no].
2. Exposed to Alcohol in the Womb [yes, no].
3. Drug dependence [none, alcohol, cocaine, heroin, other].
4. Has Ever Received Psychiatric Treatment [yes, no].
5. Treatment for alcohol or drug dependence [yes, no].
6. Anxiety or Eating Disorder [yes, no].

Which of the following best describes the main set of variables and their values discussed in the report on Fetal Exposure to Alcohol: (choose one)

A. 1, 2 and 4.

B. 2, 3 and 4.

C. 2, 4 and 6.

D. 3, 4 and 5.

Reading Exercise – No 16 (Module 4 | Causal Graphs)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

I - Reflect and discuss – Case Study 1

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6. Anxiety or Eating Disorder [yes, no].

Which of the following best describes the main set of variables and their values discussed in the report on Fetal Exposure to Alcohol: (choose one)

A. 1, 2 and 4.

B. 2, 3 and 4.

C. 2, 4 and 6.

D. 3, 4 and 5.

Answer - C.
Everyone in the study was an adult, so Adult [yes, no] is not a main variable in the study. Drug dependence was discussed, but the article made no mention of whether subjects were dependent on cocaine, heroin, etc.

Reading Exercise – No 16 (Module 4 | Causal Graphs)

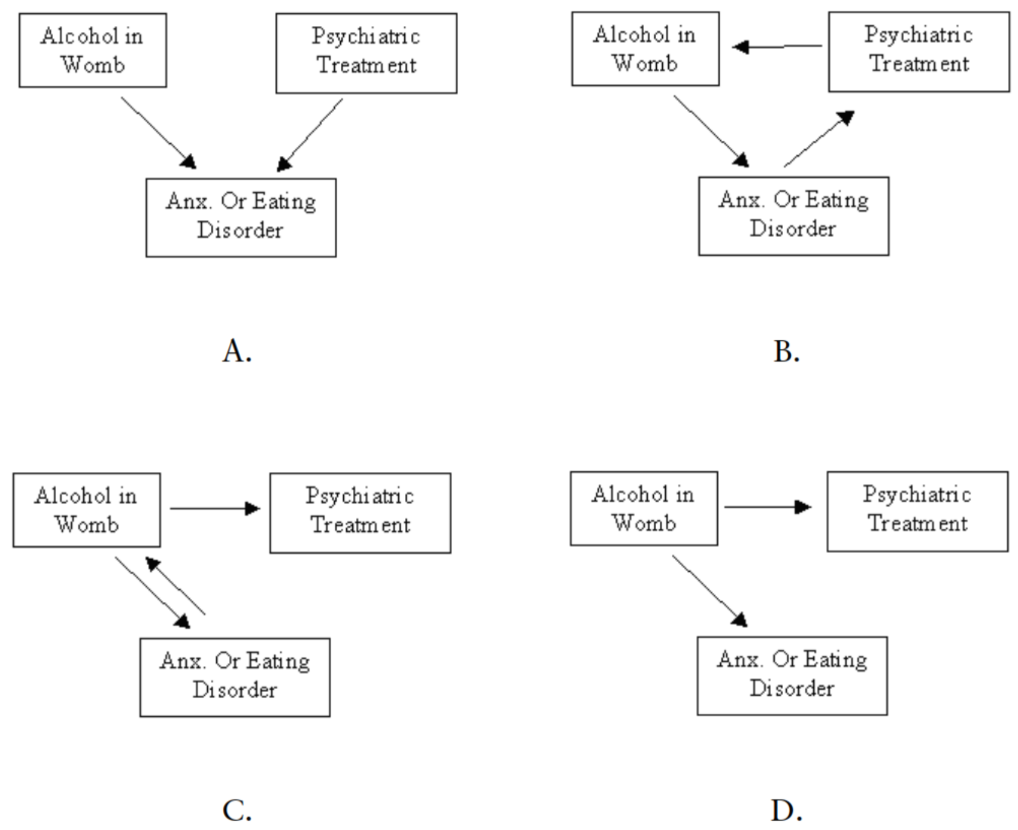
Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

I - Reflect and discuss – Case Study 1

Question No 18 – Assume that the main variables in the study were:

1. Exposed to Alcohol in the Womb [yes, no]
2. Has Ever Received Psychiatric Treatment [yes, no]
3. Anxiety or Eating Disorder [yes, no]

Which of the following is the causal graph that best captures the hypotheses discussed in the study: (Choose one)



Reading Exercise – No 16 (Module 4 | Causal Graphs)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

J - Reflect and discuss – Case Study 2

Question No 19 – Consider the following list:

1. Elderly [yes, no]
2. Exercise [3 times weekly, no]
3. Body's Blood Pumping Capacity [poor, avg., good]
4. Cognitive Function [poor, avg., good]
5. Medicated for Depression [yes, no]
6. Blood's Oxygen Carrying Capacity [poor, avg., good]

Which of the following best describes the set of variables that were measured in the report on Exercise and Mental Decline: (choose one)

A. 1, 2 and 4.
B. 2, 3 and 5.
C. 2, 4 and 6.
D. 2, 4 and 5.

Reading Exercise – No 16 (Module 4 | Causal Graphs)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

J - Reflect and discuss – Case Study 2

Question No 19 – Consider the following list:

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5. Medicated for Depression [yes, no]
6. Blood's Oxygen Carrying Capacity [poor, avg., good]

Which of the following best describes the set of variables that were measured in the report on Exercise and Mental Decline: (choose one)

A. 1, 2 and 4.
B. 2, 3 and 5.
C. 2, 4 and 6.
D. 2, 4 and 5.

Answer - D.

- Everyone in the study was elderly, so Elderly [yes, no] is not a main variable in the study.
- The body's blood pumping capacity was discussed, but only as a hypothetical link mediating Exercise and Cognitive Function. The article suggested that it was not measured in the study.
- The blood's oxygen carrying capacity was discussed, but only as a hypothetical link mediating Exercise and Cognitive Function. The article suggested that it was not measured in the study.

Reading Exercise – No 16 (Module 4 | Causal Graphs)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

J - Reflect and discuss – Case Study 2

Question No 20 – Consider again the following list:

1. Elderly [yes, no]
2. Exercise [3 times weekly, no]
3. Body's Blood Pumping Capacity [poor, avg., good]
4. Cognitive Function [poor, avg., good]
5. Medicated for Depression [yes, no]
6. Blood's Oxygen Carrying Capacity [poor, avg., good]

Which of the following best describes the set of variables that were part of a mechanism for how exercise might improve cognitive function?

A. 1, 2, 3 and 5.
B. 2, 3, 4 and 5.
C. 2, 3, 4 and 6.
D. 2, 4, 5 and 6.

Reading Exercise – No 16 (Module 4 | Causal Graphs)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

J - Reflect and discuss – Case Study 2

Question No 20 – Consider again the following list:

1. Elderly [yes, no]
2. Exercise [3 times weekly, no]
3. Body's Blood Pumping Capacity [poor, avg., good]
4. Cognitive Function [poor, avg., good]
5. Medicated for Depression [yes, no]
6. Blood's Oxygen Carrying Capacity [poor, avg., good]

Which of the following best describes the set of variables that were part of a mechanism for how exercise might improve cognitive function? (choose one)

A. 1, 2, 3 and 5.
B. 2, 3, 4 and 5.
C. 2, 3, 4 and 6.
D. 2, 4, 5 and 6.

Answer - C.

Whether or not a patient was medicated for depression was discussed, but not as a link mediating Exercise and Cognitive Function.

Reading Exercise – No 16 (Module 4 | Causal Graphs)

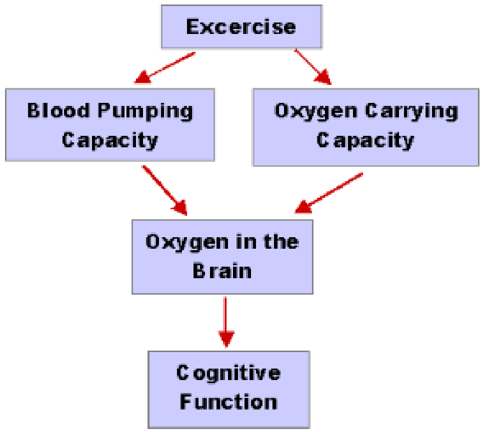
Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

J - Reflect and discuss – Case Study 2

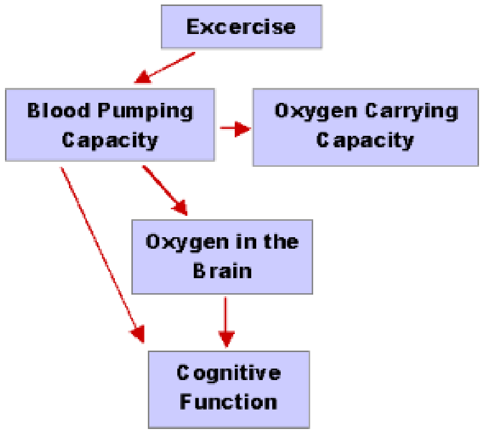
Question No 21 – Consider the following variables:

1. Exercise [3 times weekly, no]
2. Body's Blood Pumping Capacity [poor, avg., good]
3. Blood's Oxygen Carrying Capacity [poor, avg., good]
4. Oxygen Carrying Capacity [poor, avg., good]
5. Cognitive Function [poor, avg., good]

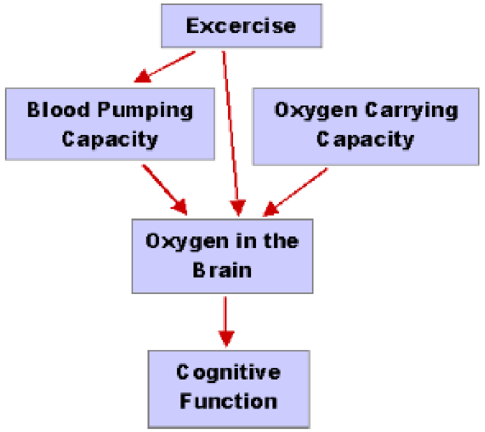
Which of the following graphs best represents the causal system consisting of the above variables? (Choose one)



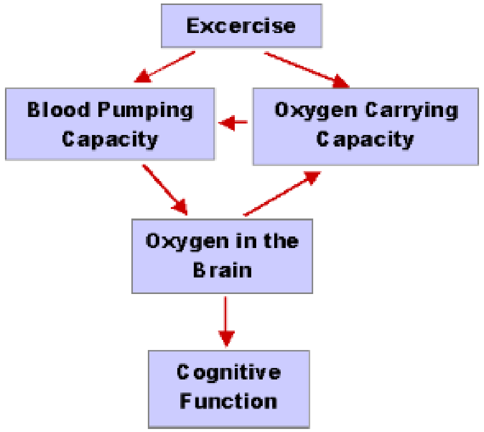
A.



B.



C.



D.

Reading Exercise – No 16 (Module 4 | Causal Graphs)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

J - Reflect and discuss – Case Study 2

Question No 21 – Consider the following variables:

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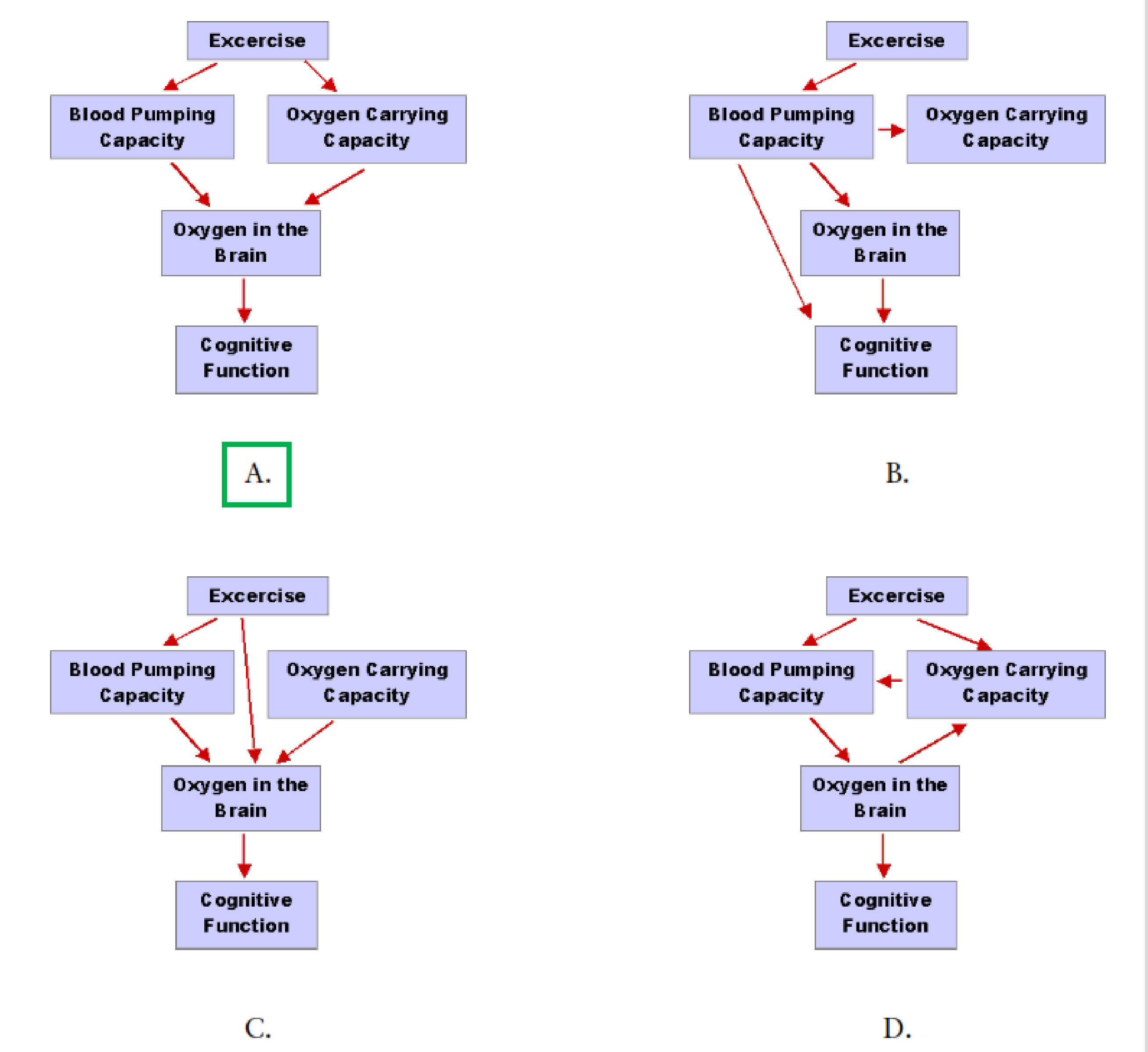

Answer - A.

- In B, Oxygen Carrying Capacity is not a cause, direct or indirect, of Cognitive Function. That contradicts the article, which suggested it was.
- In C, Exercise is not a cause of Oxygen Carrying Capacity, which the article said it was.
- In D, Oxygen to the Brain causes Oxygen Carrying Capacity, instead of vice versa.